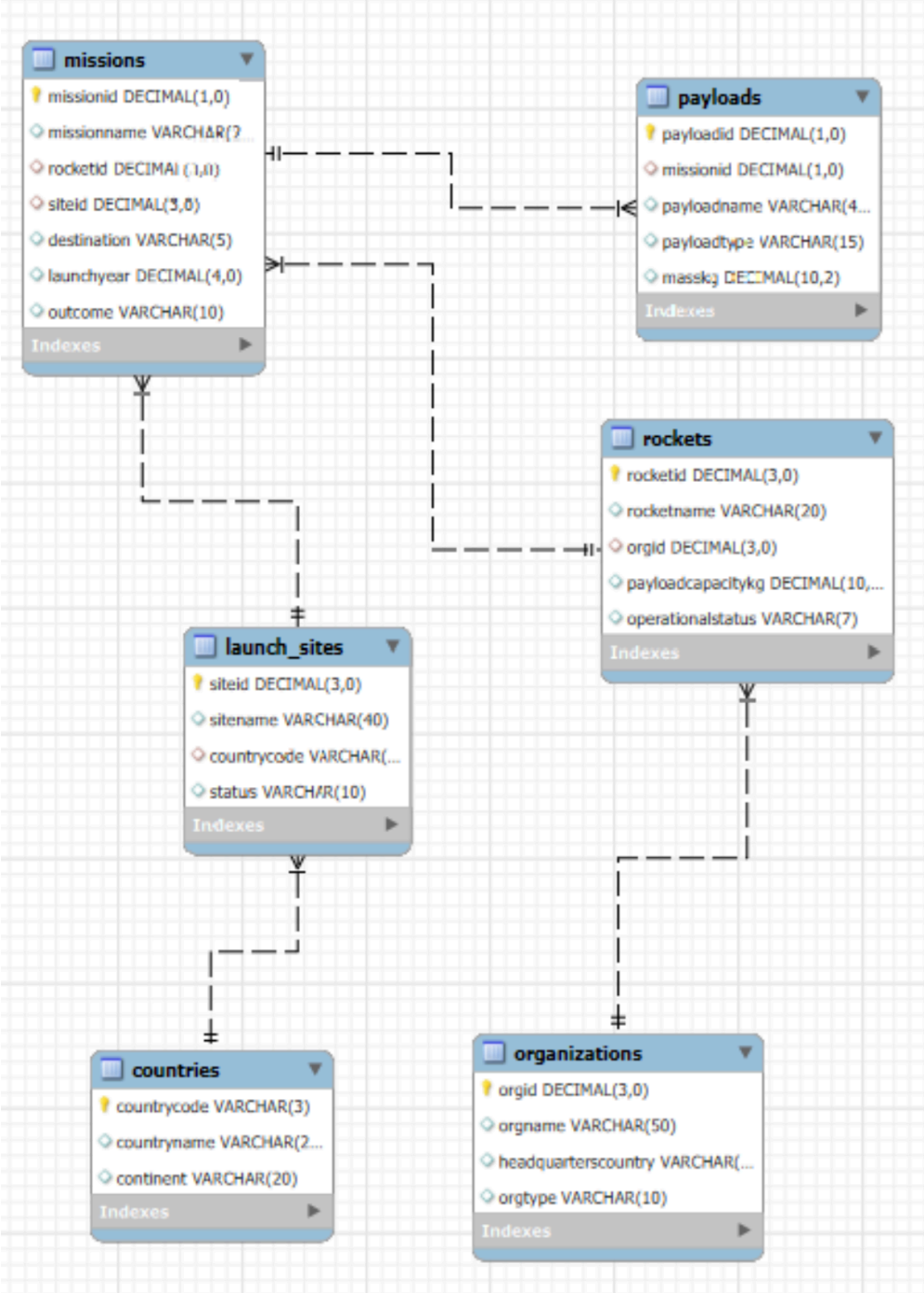
Hailey Newbold

Assignment 4

EER Diagram



1

```
107     -- 1
108 •  select missions.missionname, rockets.rocketname, organizations.orgname, launch_sites.sitename,
109            missions.destination, missions.launchyear, missions.outcome
110     from missions
111         inner join rockets on missions.rocketid = rockets.rocketid
112         inner join launch_sites on missions.siteid = launch_sites.siteid
113         inner join organizations on rockets.orgid = organizations.orgid
114     order by missions.launchyear desc, missions.missionname asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	orgname	sitename	destination	launchyear	outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

2

```
116     -- 2
117 •  select missions.missionname, rockets.rocketname, payloads.payloadname,
118            payloads.payloadtype, payloads.masskg
119     from payloads
120         inner join missions on missions.missionid = payloads.missionid
121         inner join rockets on missions.rocketid = rockets.rocketid
122     order by payloads.masskg desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800.00
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500.00
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000.00
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450.00
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200.00
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192.00
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150.00
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100.00

3

```
124     -- 3
125 •   select launch_sites.sitename, missions.missionname, rockets.rocketname, missions.launchyear
126     from launch_sites
127         left join missions on launch_sites.siteid = missions.siteid
128         left join rockets on missions.rocketid = rockets.rocketid
129     order by launch_sites.sitename asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

4

```
131     -- 4
132 •   with payloadmass as (
133         select organizations.orgname, organizations.headquarterscountry, organizations.orgtype,
134                 sum(payloads.masskg) as total
135         from organizations
136             inner join rockets on organizations.orgid = rockets.orgid
137             inner join missions on missions.rocketid = rockets.rocketid
138             inner join payloads on payloads.missionid = missions.missionid
139         group by organizations.orgname, organizations.headquarterscountry, organizations.orgtype
140     )
141     select orgname, headquarterscountry, orgtype, total
142     from payloadmass
143     order by total desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orgname	headquarterscountry	orgtype	total
National Aeronautics and Space Administration	USA	Government	25800.00
Space Exploration Technologies Corp	USA	Private	13692.00
Japan Aerospace Exploration Agency	JPN	Government	6000.00
Roscosmos State Corporation	RUS	Government	2450.00
China National Space Administration	CHN	Government	1200.00
Indian Space Research Organisation	IND	Government	150.00
Rocket Lab	NZL	Private	100.00

5

```sql
-- 5
with missionsbycc as (
    select launch_sites.countrycode, sum(missions.missionid) as totalmissions
    from missions
        inner join launch_sites on missions.siteid = launch_sites.siteid
    group by launch_sites.countrycode
),
payloadmass as (
    select launch_sites.countrycode, sum(payloads.masskg) as totalpayload
    from payloads
        inner join missions on missions.missionid = payloads.missionid
        inner join launch_sites on missions.siteid = launch_sites.siteid
    group by launch_sites.countrycode
)
select countries.countryname, totalmissions, totalpayload
from countries
    inner join missionsbycc on missionsbycc.countrycode = countries.countrycode
    inner join payloadmass on payloadmass.countrycode = countries.countrycode
group by countries.countrycode
order by totalpayload desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

countryname	totalmissions	totalpayload
United States	11	39492.00
Japan	6	6000.00
Russia	5	2450.00
China	4	1200.00
India	3	150.00
New Zealand	7	100.00